



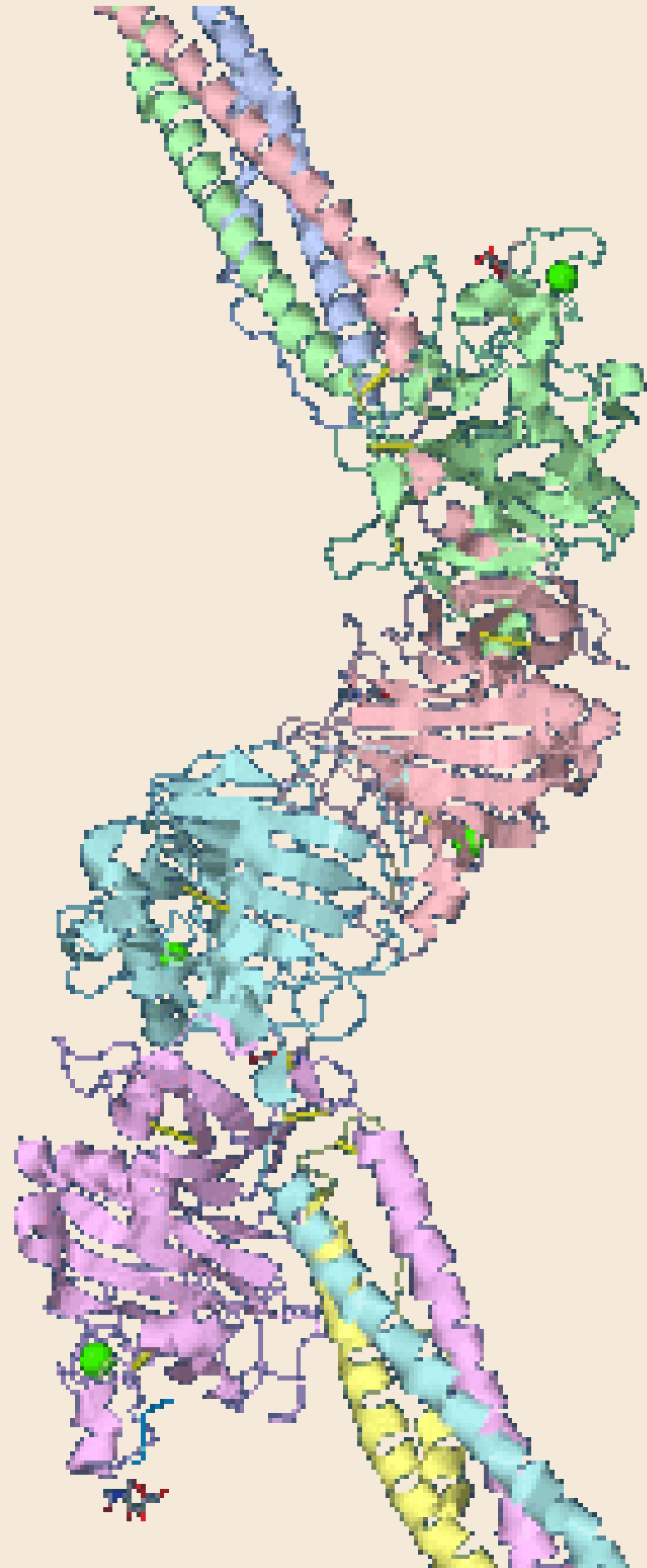
# HUMAN FIBRIN : HOW PROTEIN ARCHITECTURE CONTROLS BLOOD CLOTTING



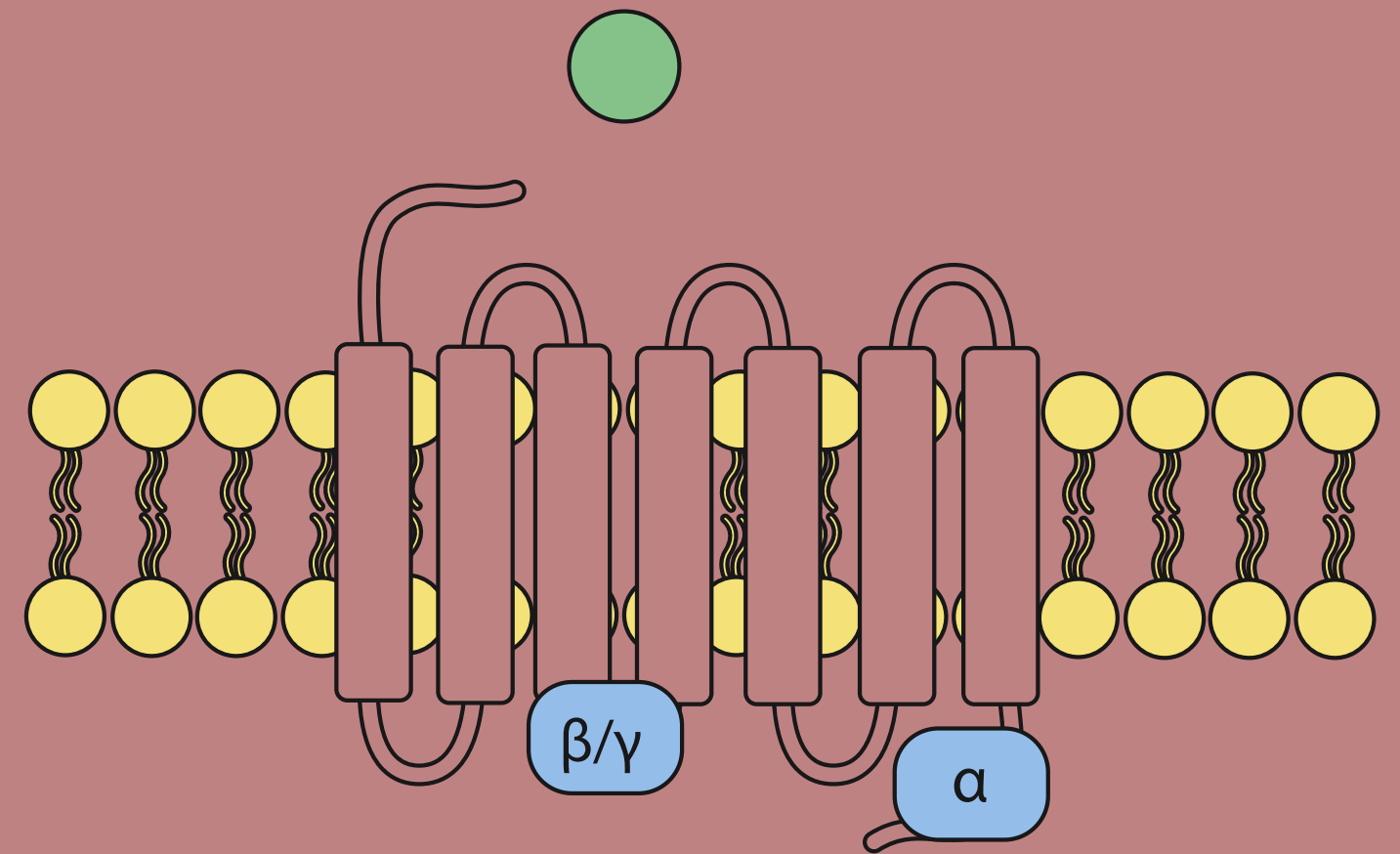
BY GROUP B



SO, WHAT DOES IT REALLY DO?!



# HUMAN FIBRIN



# CAUTION

**HUMAN FIBRIN IS NOT JUST A CLOTTING  
PROTEIN; IT IS A STRUCTURAL PLATFORM  
THAT DETERMINES WHETHER BLOOD  
LOSS IS STOPPED OR LIFE IS  
THREATENED**



# FROM CLOT TO CURE: FIBRIN TURNS INJURY INTO REPAIR.



# PHYSIOLOGICAL ROLES OF FIBRIN

GENERATED FROM SOLUBLE  
FIBRINOGEN DURING THE  
COAGULATION CASCADE.

STRICTLY REGULATED TO AVOID  
PATHOLOGICAL CLOTTING

FORMS THE STRUCTURAL  
FRAMEWORK OF BLOOD CLOTS TO  
PREVENT EXCESSIVE BLEEDING

BLOOD VESSEL DAMAGE ACTIVATES  
ENZYMES, LEADING TO THROMBIN.

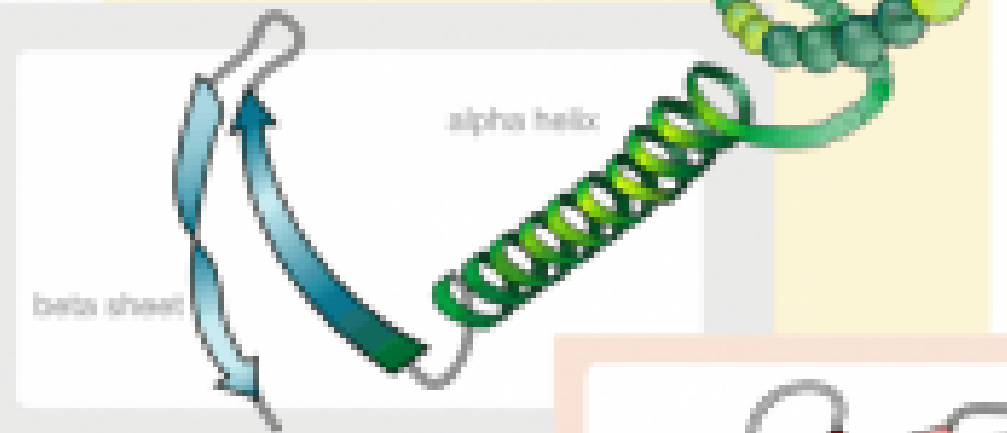
THROMBIN CONVERTS FIBRINOGEN  
INTO FIBRIN AT THE INJURY SITE.

FIBRIN STABILIZES CLOTS AND  
SUPPORTS TISSUE REPAIR IN A  
TIGHTLY REGULATED MANNER

SO, WHAT ARE THE STRUCTURAL  
LEVELS OF THIS PROTEIN?!!



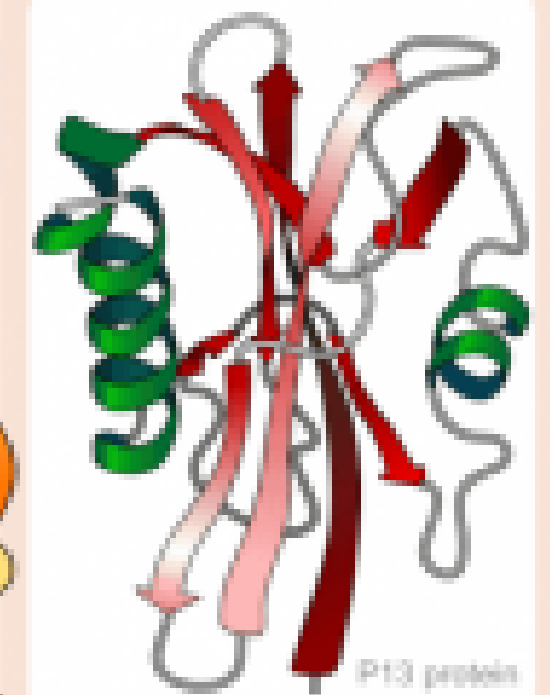
Primary structure  
amino acid sequence



Secondary structure  
regular sub-structures



Quaternary structure  
complex of protein molecules



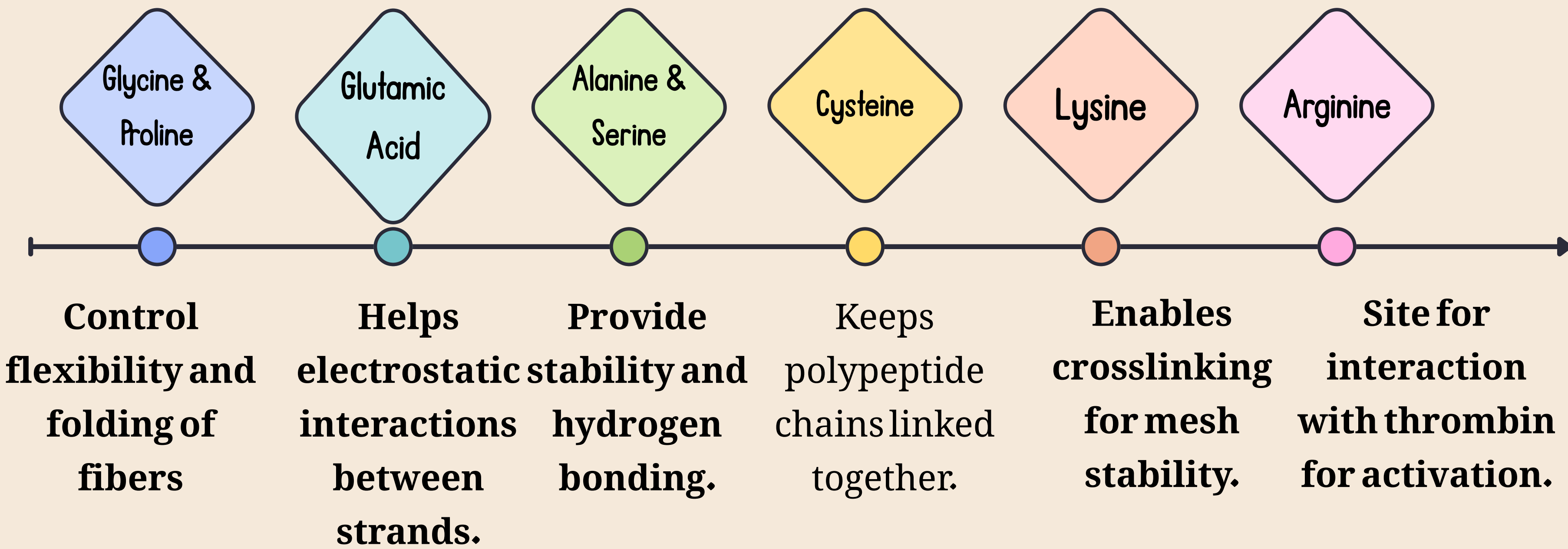
Tertiary structure  
three-dimensional structure

# PRIMARY STRUCTURE OF FIBRIN



**POLYPEPTIDE CHAINS: COMPOSED OF THREE TYPES →  $A\alpha$ ,  $B\beta$ ,  $\gamma$ .**  
**EACH CHAIN IS PRESENT IN TWO IDENTICAL COPIES → STRUCTURE:  $(A\alpha)_2(B\beta)_2(\gamma)_2$ .**

**HEPTAD REPEAT:**  
**A REPEATING SEVEN-AMINO ACID PATTERN IN A PROTEIN SEQUENCE THAT PROMOTES ALPHA-HELIX AND COILED-COIL FORMATION**



# STRUCTURE-FUNCTION RELATIONSHIP

**FIBRINOPEPTIDES: SPECIFIC SHORT SEQUENCES (FIBRINOPEPTIDE A AND B) ON THE  $A\alpha$  AND  $B\beta$  CHAINS. CRUCIAL ROLE: THESE REGIONS CONTAIN KEY FUNCTIONAL SITES.**

01

**Binding sites for fibrin chain assembly.**

02

**Thrombin interaction sites for activation.**

03

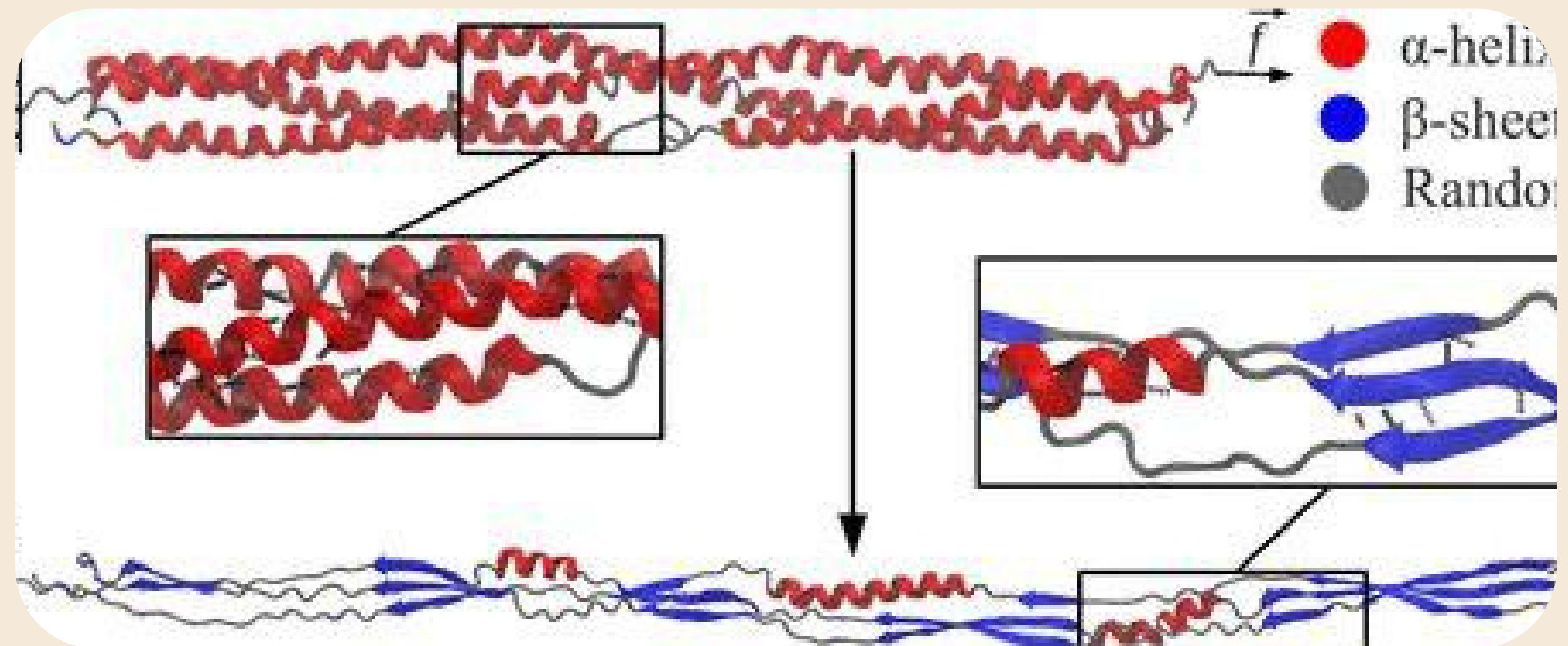
**ESSENTIAL FOR FORMING THE FIBRIN MESH TO STOP BLEEDING.**

**THE KEY EVENT: THROMBIN CLEAVES OFF THE FIBRINOPEPTIDES A AND B. CLINICAL IMPACT: MUTATIONS IN THESE SEQUENCES CAN CAUSE DEFECTIVE CLOTTING AND BLEEDING DISORDERS.**

# SECONDARY STRUCTURE OF FIBRIN



**FIRST LEVEL  
OF FOLDING**



**DETERMINING LOCAL  
CONFORMATION**

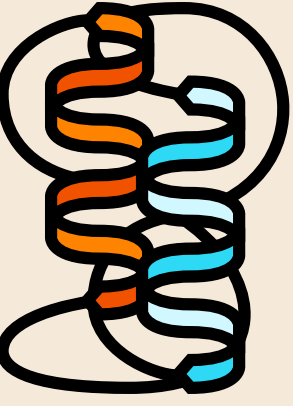
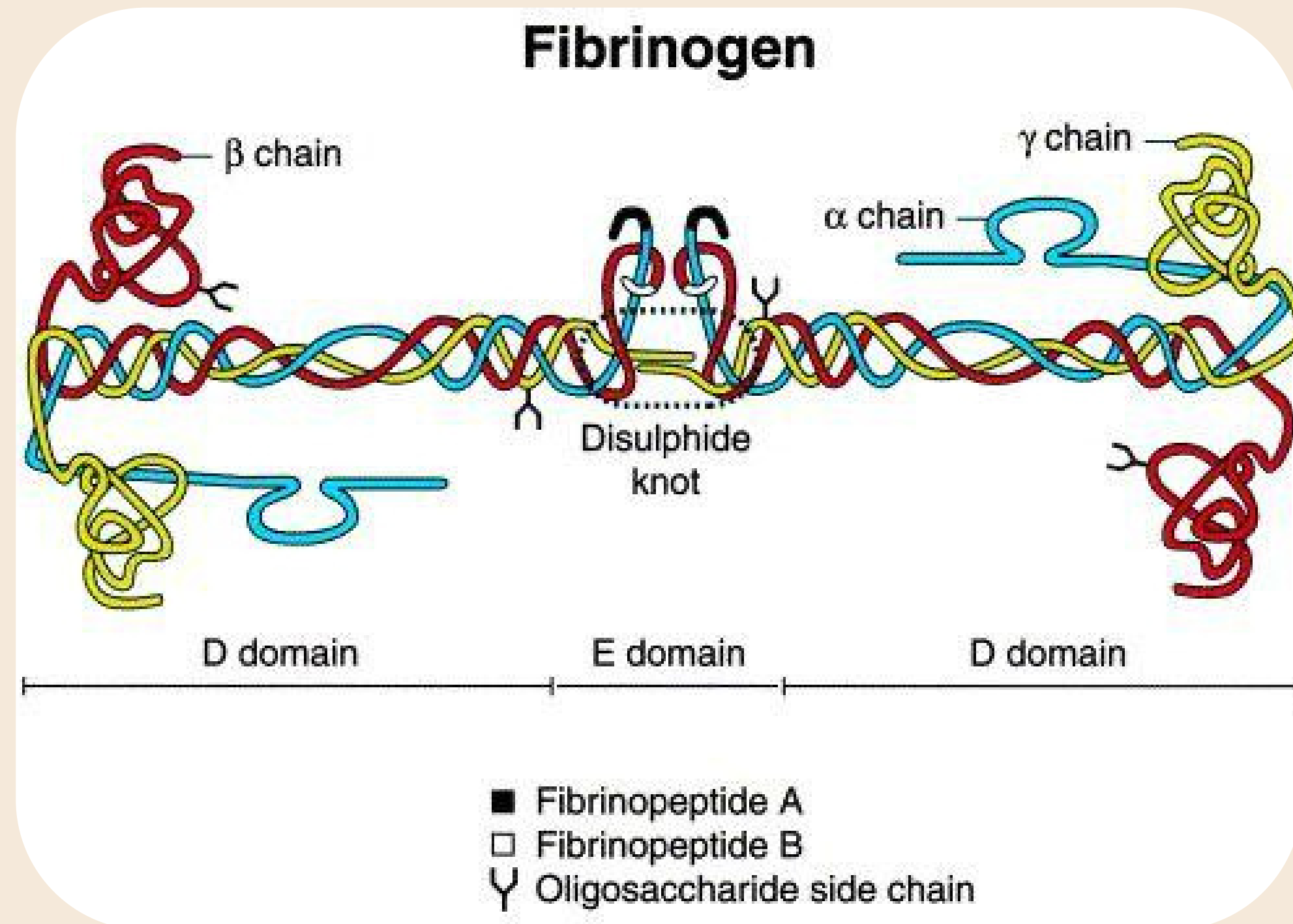
**ADOPTION OF ALPHA  
HELIX AND BETA SHEET**



**FIBRINOPEPTIDES  
AND CHAIN  
FEATUERS**

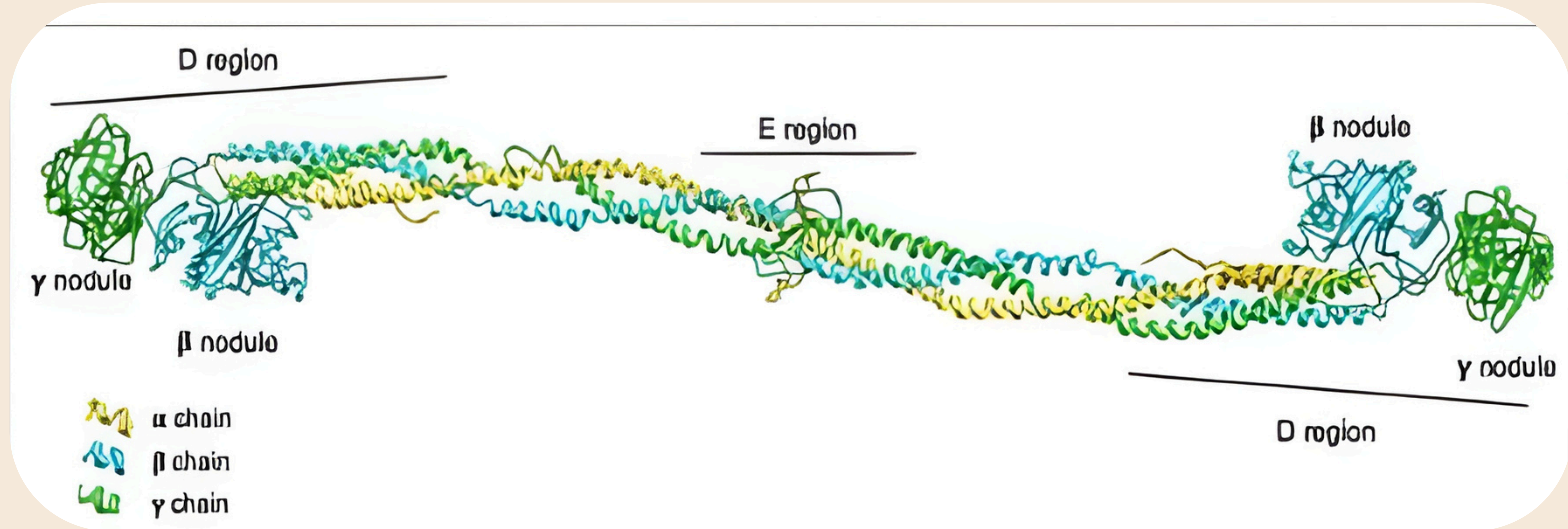
**ALPHA HELICES  
AND COILED-COIL MOTIFS**

**THE ALPHA C  
REGION**



# STRUCTURAL-FUNCTIONAL RELATIONSHIP

2  
LEVEL



## COILED-COIL MOTIFS

PROVIDE CONNECTIVITY  
BETWEEN DOMAINS

## ALPHA-C REGION

FLEXIBLE REGION FACILITATING  
POLYMERIZATION AND CROSS-  
LINKING"

## LOCALIZED B-SHEETS

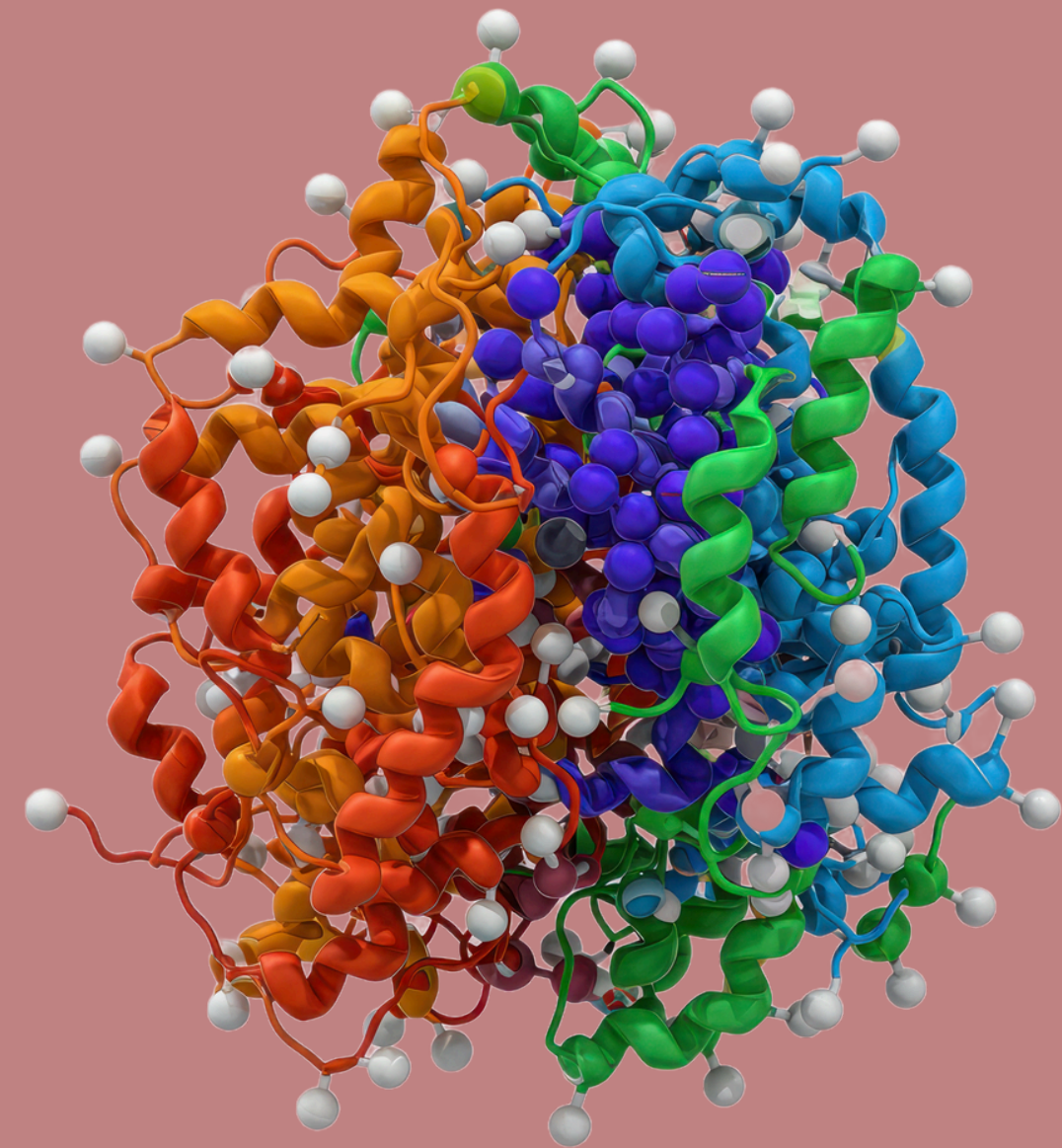
ENHANCE LOCAL STRUCTURAL  
STABILITY AND HELP  
PREVENT TEARING

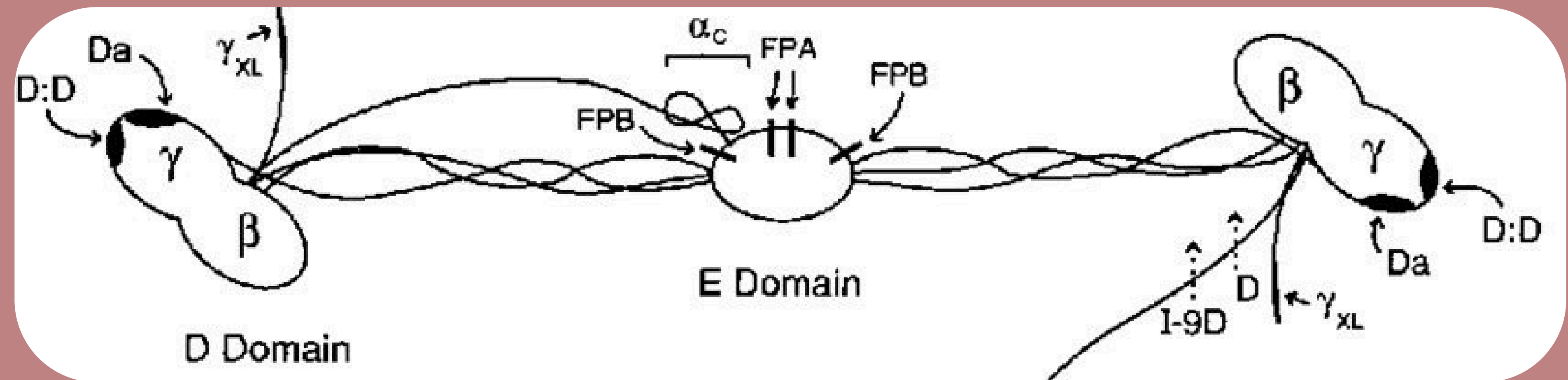
# TERTIARY STRUCTURE OF FIBRIN

**IT IS THE 3D ARRANGEMENT OF FIBRIN'S POLYPEPTIDE CHAINS.**

**FIBRIN IS A FIBROUS PROTEIN → ELONGATED AND EXTENDED SHAPE (UNLIKE GLOBULAR PROTEINS).**

**THIS STRUCTURE IS CRITICAL FOR BLOOD CLOT FORMATION, PROVIDING MECHANICAL STRENGTH AND FLEXIBILITY.**





# COMPOSITION & STRUCTURAL FEATURES



**DERIVED FROM FIBRINOGEN AFTER CLEAVAGE BY THROMBIN.**

**FIBRIN MONOMER COMPOSITION:**

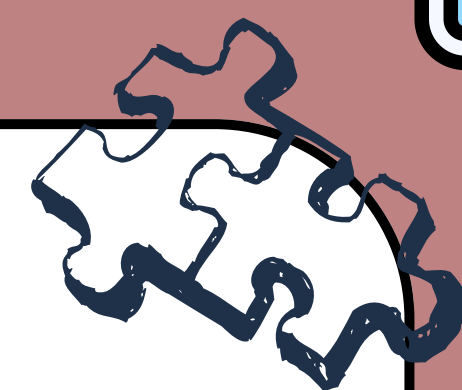
**2 A-CHAINS**

**2 B-CHAINS**

**2  $\Gamma$ -CHAINS**

**CHAINS FOLD INTO A-HELICES AND B-SHEETS, ARRANGED IN AN EXTENDED SHAPE.**

**E AND D DOMAINS HELP IN POLYMERIZATION AND BLOOD CLOTTING**

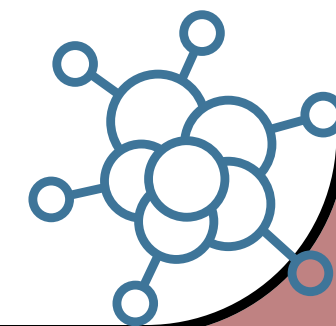


**COILED-COIL CONNECTORS:**

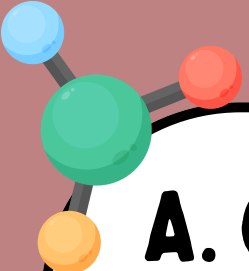
**CENTRAL PART OF MONOMER FORMED BY A-HELICES TWISTED TOGETHER → COILED-COIL MOTIF.**

**CONNECTS GLOBULAR DOMAINS AT ENDS OF MONOMER (D AND E REGIONS).**

**PROVIDES STRUCTURAL RIGIDITY AND ENSURES PROPER ALIGNMENT FOR POLYMERIZATION.**



# BONDS & INTERACTIONS STABILIZING TERTIARY STRUCTURE

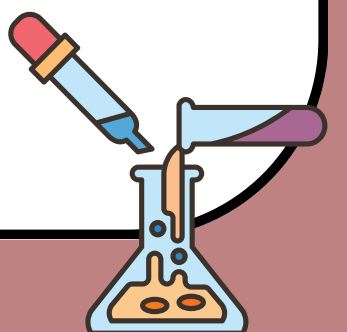
A small molecular model with a central green sphere and three surrounding spheres in blue, red, and orange.

## A. COVALENT BONDS

- **DISULFIDE BRIDGES (S-S BONDS) → STABILIZE POLYPEPTIDE CHAINS.**

## B. NONCOVALENT INTERACTIONS

- **HYDROGEN BONDS → MAINTAIN A-HELICES AND B-SHEETS (INCLUDING COILED-COIL HELICES).**
- **IONIC BONDS (SALT BRIDGES) → STABILIZE CHARGED RESIDUES.**
- **HYDROPHOBIC INTERACTIONS → KEEP NONPOLAR RESIDUES INSIDE COILED-COIL CORE.**
- **VAN DER WAALS FORCES → STABILIZE CLOSELY PACKED REGIONS.**



# STRUCTURAL–FUNCTIONAL RELATIONSHIP

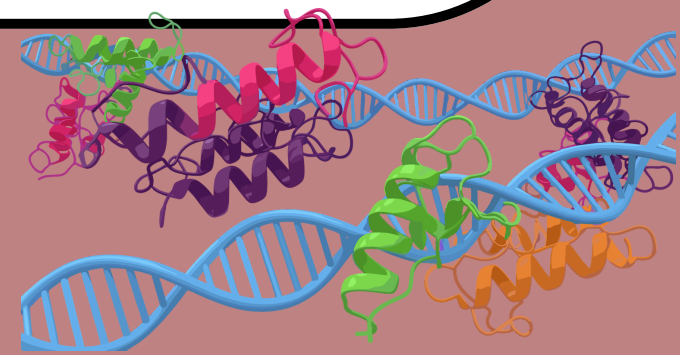


**EXTENDED SHAPE + COILED-COIL CONNECTORS → ALLOW POLYMERIZATION INTO LONG FIBERS → FORMS STABLE CLOT NETWORK.**

**COILED-COIL REGIONS ENSURE CORRECT ALIGNMENT & SPACING BETWEEN GLOBULAR DOMAINS → CRITICAL FOR FIBER FORMATION.**

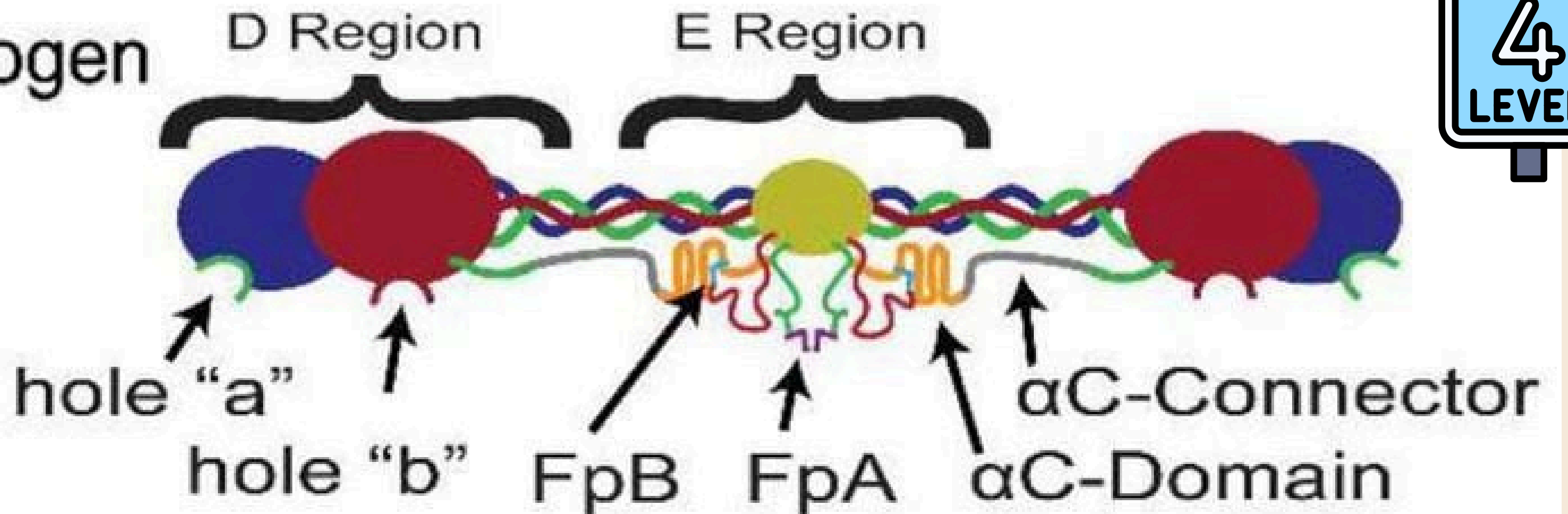
**MECHANICAL STRENGTH & ELASTICITY COME FROM COMBINATION OF EXTENDED TERTIARY STRUCTURE AND COILED-COIL MOTIFS.**

**E AND D DOMAINS → POLYMERIZATION.**



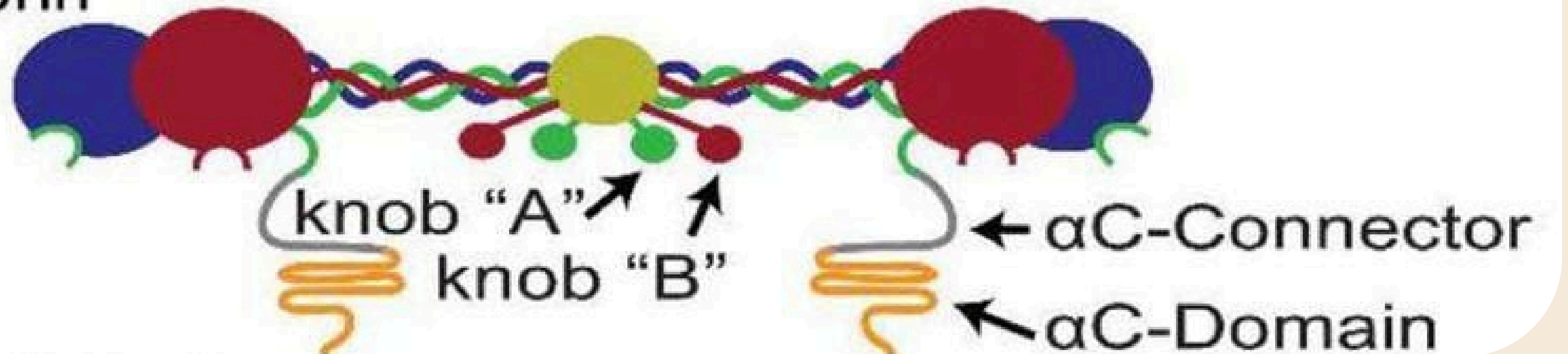
(a) Fibrinogen

5 nm



(b) Fibrin

5 nm



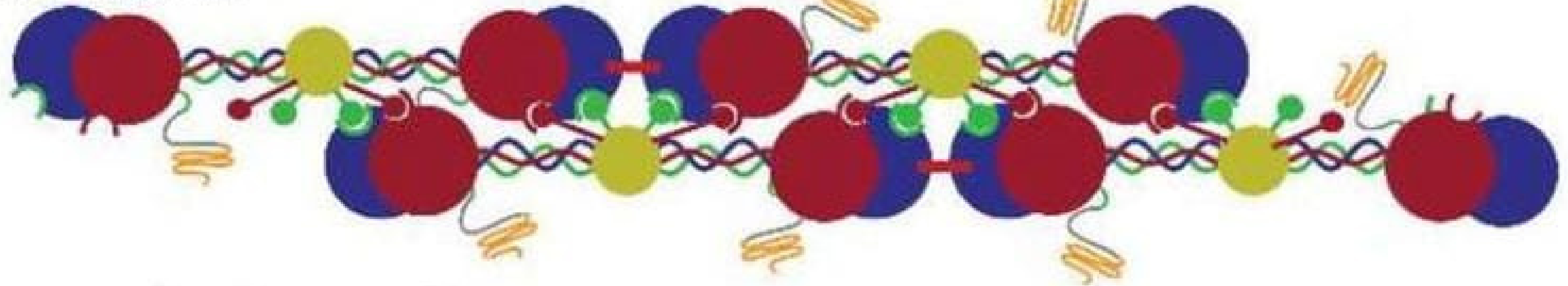
**A) START OF QUATERNARY STRUCTURE**

**B) KNOB-HOLE INTERACTION**

# PROTOFIBRIL FORMATION

(c) Protofibril

10 nm



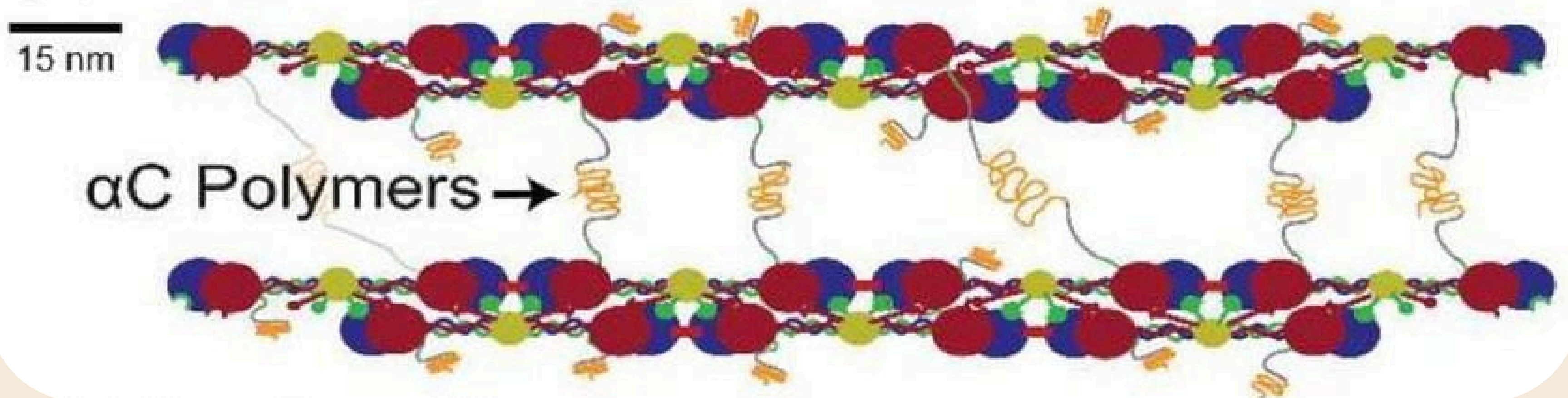
 A:a or B:b  
knob:hole interaction

  $\gamma$ — $\gamma$  interaction

# LATERAL AGGREGATION

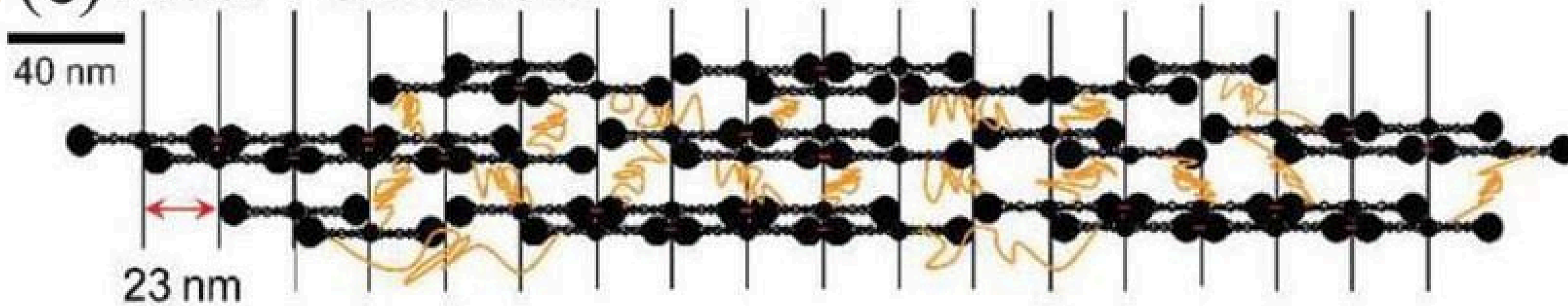


(d) Protofibril Bundling

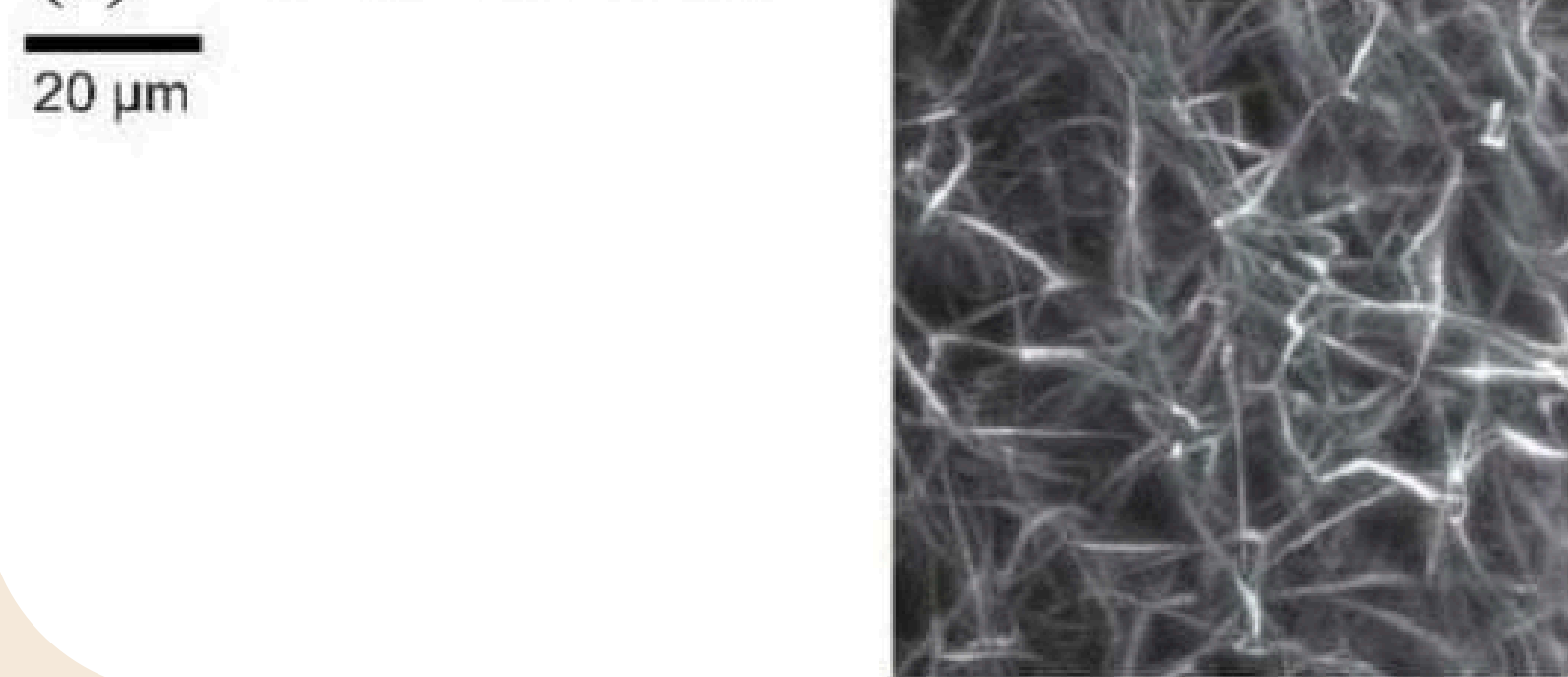


# CROSS-LINKING AND STABILIZATION

(e) Fiber Formation



(f) Fibrin Network



**FACTOR XIII**

# DO YOU HAVE ANY QUESTIONS?



# PREPARED & PRESENTED BY



## A GIRL

**NADA ANWAR**

**MAHA HOSSAM**

**ALYAA EKRAMY**

**DOAA ABDELFATTAH**

**MEHRAEIL FAYEK**

**YASMINE ABDELRAOUF**

**HADEER HANY**

**BASMALA NASSER**

**DOAA MARAWAN**

